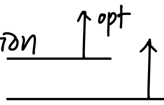
Min $f(x_1,x_2)=c_1x_1+c_2x_2$ 하려면 $-(c_1,c_2)$ 방향으로 움직이기.
↳ gradient의 반대방향.

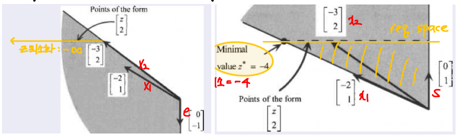
LP in terms of solution

1. unique optimal solution
2. alternative optimal solution
3. unbounded
4. infeasible



min
$(z_j-c_j<0)$
$(z_j-c_j=0)$
$(cd<0)$

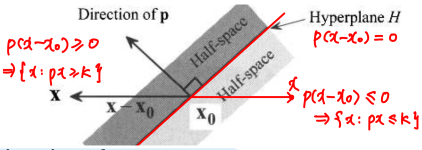
Requirement Space : RHS(b)의 space → a_j의 span 안에 b가 있어야 feas.
↳ opt 관점 : $\sum_{j=1}^{n}\begin{bmatrix}c_j\\a_j\end{bmatrix}x_j=\begin{bmatrix}z\\b\end{bmatrix}$



Linear Independence
$\sum\lambda_ja_j=0$ implies that $\lambda_j=0\ \forall j$
Basis : 하나의 a_j 라도 빠지면 $\mathbb{R}^n$ span 불가. / rank : # of indep. row/col.
rank(A,b) > rank(A) : no sol / = : unique sol / < : inf. sol.

Convex Sets → hyperplane = convex set

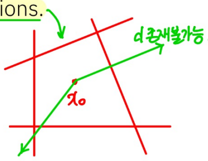
- A set $\mathbf{X}\in\mathbb{R}^n$ is called a convex set if given any two points $x_1,x_2\in\mathbf{X}$, then $\lambda\mathbf{x}_1+(1-\lambda)\mathbf{x}_2\in\mathbf{X}$ for each $\lambda\in[0,1]$.
- Geometrically, any line segment connecting $\mathbf{x}_1,\mathbf{x}_2\in\mathbf{X}$ must belong to $\mathbf{X}$.



min cx. s.t. $\underbrace{Ax\leq b}_{m},\ \underbrace{x\geq 0}_{n}$
- hyperplane m+n 개
- intersection of (m+n) halfspaces.

Direction of a convex set.

- Given a convex set, a nonzero vector $\mathbf{d}$ is called (recession) direction of the set, if for each $\mathbf{x}_0$ in the set, the ray $\{\mathbf{x}_0+\lambda\mathbf{d}:\lambda\geq 0\}$ also belongs to the set.
  - Clearly, if the set is bounded, then it has no directions.
- The set of directions of $\mathbf{X}$ is called the recession cone.
  - The recession cone of $\mathbf{X}=\{\mathbf{x}:\mathbf{Ax}\leq\mathbf{b},\mathbf{x}\geq 0\}$ is $D=\{\mathbf{d}:\underbrace{\mathbf{Ad}\leq 0}_{①},\underbrace{\mathbf{d}\geq 0}_{②},\mathbf{d}\neq 0\}$.
    영이면 안되니까



$X:\{x: Ax\leq b,\ x\geq 0\}$
$R:\{x: x=x_0+\lambda d,\ \lambda\geq 0\}$ $Ax=A(x_0+\lambda d)=Ax_0+\lambda Ad\leq b$
$Ax_0\leq b$ 이에 $\lambda Ad\leq 0$ 이어야함
$\lambda\geq 0$ 이니까 $Ad\leq 0$ ① $x=x_0+\lambda d\geq 0.\ \lambda d\geq 0.\ d\geq 0$ ②

무수히 많은 recession 중, 대표하는 ED → $1d=1$ 로 normalize 해서 구하기.
↳ D 차원에서 feasible 한지 확인해봐야함!!!

Convex Cone $C=\{\sum_{j=1}^{k}\lambda_ja_j:\lambda_j\geq 0,\ \forall j\}$

- A function f of the vector of variables $(x_1,x_2,\dots,x_n)$ is said to be convex if the following inequality holds for any two vectors x_1 and x_2:
$$f(\lambda x_1+(1-\lambda)x_2)\leq\lambda f(x_1)+(1-\lambda)f(x_2),\ \text{for all }\lambda\in[0,1]$$
- A function f of the vector of variables $(x_1,x_2,\dots,x_n)$ is said to be concave if the following inequality holds for any two vectors x_1 and x_2:
$$f(\lambda x_1+(1-\lambda)x_2)\geq\lambda f(x_1)+(1-\lambda)f(x_2),\ \text{for all }\lambda\in[0,1]$$

A polyhedral set or a polyhedron is the intersection of a finite number of half-spaces. *degenerate 이어도 X를 형성하는데 쓰이면 defining hyperplane

- A bounded polyhedral set is called a polytope. $\{x: Ax\leq b\}$

A polyhedral cone is the intersection of a finite number of half-spaces, whose hyperplanes pass through the origin. $\{x: Ax\leq 0\}$

- ★ We call the hyperplanes associated with the $(m+n)$ halfspaces defining $\mathbf{X}$, the *defining hyperplanes*.
- We call a set of hyperplanes *linearly independent* if the *coefficient matrix* associated with the is *full rank*, i.e., all its rows are linearly independent.

Alternative definition of extreme point

- $\bar{x}\in\mathbf{X}$ is an extreme point of $\mathbf{X}$ if it lies on some n linearly independent defining hyperplanes of $\mathbf{X}$. n개보다 많은만큼 order of degeneracy
- The two definitions of extreme point are equivalent (prove!).
  1. $\bar{x}$ lies on n linearly independent defining hyperplane.
  2. It cannot be written as a strict convex combination of two distinct points in $\mathbf{X}$. → $x_1=x_2=x$ 일수밖에 없음

Face 전체 공간의 차원

- In general the set of points in $\mathbf{X}$ that lie on a nonempty subset of defining hyperplanes of $\mathbf{X}$ is called a *face* of $\mathbf{X}$. (n linearly indep. hyperplane)
- $r(F)$ = max number of linearly independent defining hyperplanes satisfied at all points of F.
- *Dimension* of face F: $\dim(F)=n-r(F)$
- Extreme points are *zero-dimensional* faces.
- A one dimensional face is called an *edge*.

• adjacent E.P.
: (n−1) ind. linear hyperplanes in common

Representation Theorem → polyhedron은 ED와 EP로 나타낼수있다.

Let $\mathbf{X}=\{\mathbf{x}:\mathbf{Ax}\leq\mathbf{b},\mathbf{x}\geq 0\}$ be a nonempty (polyhedral) set.

- Then the set of extreme points is nonempty and has a finite number of elements, say $x_1,x_2,\dots,x_k$.
- Furthermore, the set of extreme directions is empty if and only if $\mathbf{X}$ is bounded.
- If $\mathbf{X}$ is not bounded, then the set of extreme directions is nonempty and has a finite number of elements, say $d_1,d_2,\dots,d_\ell$.
- Moreover, $\bar{x}\in\mathbf{X}$ if and only if it can be represented as a convex combination of $x_1,\dots,x_k$ plus a nonnegative linear combination of $d_1,\dots,d_\ell$, that is,

$$\bar{x}=\sum_{j=1}^{k}\lambda_jx_j+\sum_{j=1}^{\ell}\mu_jd_j$$
$$\sum_{j=1}^{k}\lambda_j=1$$
$$\lambda_j\geq 0\quad\forall j=1,\dots,k$$
$$\mu_j\geq 0\quad\forall j=1,\dots,\ell.$$

convex comb. of EPs
non negative combination of EDs

대입하면

Minimize $\mathbf{cx}$
subject to $\mathbf{Ax}=\mathbf{b}$
$\mathbf{x}\geq 0.$

Minimize $\sum_{j=1}^{k}(\mathbf{c}x_j)\lambda_j+\sum_{j=1}^{\ell}(\mathbf{c}d_j)\mu_j$
subject to $\sum_{j=1}^{k}\lambda_j=1$
$\lambda_j\geq 0\quad\forall j=1,\dots,k$
$\mu_j\geq 0\quad\forall j=1,\dots,\ell.$

λ_j,μ_j 문제 ⟹ 변환

- Since the μ_j-variables can be made arbitrarily large, the minimum is $-\infty$ if $\mathbf{c}d_j<0$ for some $j\in\{1,\dots,\ell\}$. : unbounded
- If $\mathbf{c}d_j\geq 0$ for all $j=1,\dots,\ell$, then the corresponding μ_j can be chosen as zero.
- Now, in order to minimize $\sum_{j=1}^{k}(\mathbf{c}x_j)\lambda_j$ over $\lambda_1,\dots,\lambda_k$ satisfying $\lambda_j\geq 0$ for $j=1,\dots,k$, and $\sum_{j=1}^{k}\lambda_j=1$.
- We simply find the minimum $\mathbf{c}x_j$, say $\mathbf{c}x_p$, let $\lambda_p=1$, and set all other λ_j-variables equal to zero. → bounded 라면 EP중에 최적해 존재한다 증명.

optimal E.P. → opt. sol x^*
↕ degeneracy 가 없을 때.
BFS (Basic Feasible Sol)

BFS → ~~E.P~~ → x^*
생략가능.

EP가 more than one basis를 가지면 degenerate.

LP : min $z=cx$
s.t. $Ax=b$ b는 양수로 세팅

→ $(B,N)\begin{pmatrix}x_B\\x_N\end{pmatrix}=b.\ Bx_B+Nx_N=b$
$x_B=B^{-1}b-B^{-1}Nx_N$ → BS ($_nC_m$개)
$x_B\geq 0$: BFS

$z=cx=(c_B,c_N)\begin{pmatrix}x_B\\x_N\end{pmatrix}=c_Bx_B+c_Nx_N$
$=c_B(B^{-1}b-B^{-1}Nx_N)+c_Nx_N$
$=c_B(B^{-1}b-\sum_{j\in J}B^{-1}a_jx_j)+\sum_{j\in J}c_jx_j$
$=c_BB^{-1}b-\sum_{j\in J}(c_BB^{-1}a_j-c_j)x_j$

목적식 : $z=z_0-\sum_{j\in J}(z_j-c_j)x_j$
제약식 : $B^{-1}Nx_N+x_B=B^{-1}b$
$\sum_{j\in J}B^{-1}a_jx_j+x_B=B^{-1}b$
$x_B\geq 0,\ x_j\geq 0,\ \forall j\in J$

BFS : $x_B,\ x_N$
1개씩 swap : adjacent

$z_j-c_j>0$ 면 x_k를 증가시키면 z↓
$z_j-c_j<0$ 면 x_k를 증가시키면 z↑
$z_j-c_j=0$ 면 x_k는 증가시켜도 z 유지

nbv 문제로 변환
min $z=c_Bx_B-\sum_{j\in J}\underbrace{(c_BB^{-1}a_j-c_j)}_{\text{reduced cost}}x_j$
s.t. $\sum_{j\in J}B^{-1}a_jx_j+x_B=B^{-1}b$
$x_B=\underbrace{B^{-1}b}_{=\bar{b}}-\sum_{j\in J}\underbrace{B^{-1}a_j}_{=y_j}x_j$

x_k

Opt. Condition : $z_j-c_j\leq 0$
if not. 즉, $z_j-c_j>0$ 이면,
entering var x_k는
$z_k-c_k=\max_{j\in J}\{(z_j-c_j),0\}$로 결정

$$\rightarrow\begin{bmatrix}x_{B1}\\x_{B2}\\\vdots\\\boxed{x_{Br}}\\\vdots\\x_{Bm}\end{bmatrix}=\begin{bmatrix}\bar{b}_1\\\bar{b}_2\\\vdots\\\boxed{\bar{b}_r}\\\vdots\\\bar{b}_m\end{bmatrix}-\begin{bmatrix}y_{1k}\\y_{2k}\\\vdots\\\boxed{y_{rk}}\\\vdots\\y_{mk}\end{bmatrix}x_k=0$$

leave basis

[if $y_{ik}>0$ then x_{Bi}↓]
if $y_{ik}<0$ then x_{Bi}↑
if $y_{ik}=0$ then x_{Bi} stay

Feas. Cond. : $x_k=\frac{\bar{b}_r}{y_{rk}}=\min\{\frac{\bar{b}_i}{y_{ik}}:y_{ik}>0\}$

Simplex Method (Min/Max 조심)
① constraint equation form으로 만들기
↳ b 양수유지 (+s, −e+a 중에 유동적으로 결정)
② $A, B, B^{-1}, c_B, c_BB^{-1}, b, B^{-1}b, z_j-c_j$ → entering x_k 결정
③ $x_B=B^{-1}b-B^{-1}a_kx_k$ 에서 $y_k=B^{-1}a_k$ 활용해서 leave 결정

[Alternative Optimal Solution]
$z_j-c_j=0$ 이 있으면 iteration 더해서 E.P 구해서 generalized form 으로.
$X=\{(x)\mid(x)=\lambda_1(x_1)+\lambda_2(x_2),\ \lambda_1+\lambda_2=1,\ \lambda_1,\lambda_2\geq 0\}$

[Unboundedness] $z_k-c_k>0$ 인데 $y_k\leq 0\ (\forall k)$ 라면!
* NBV 후보 중에 하나라도 모든 y가 ≤ 0 이면 unbounded.
$[x_{BV}]=[B^{-1}b]-[y_k]x_k=[B^{-1}b-y_k\cdot x_k]$
$d=[x]$ · x_k가 1만큼 증가하면 나머지가 어떻게 변하는지. → 부호건사
unboundness 검증 : $cd<0$

[Simplex Tableau]

	z	x_B	x_N	RHS
z	1	0	$C_B B^{-1}N - C_N$ $= z_j - c_j$	$C_B B^{-1} b$
x_B	0	I	$B^{-1}N$ $= y_j$	$B^{-1}b$

$[B^{-1}]$

$z_k - c_k > 0$ and $y_{rk} > 0$.

$(z_{Br} - c_{Br})_{new}$

$= (z_{Br} - c_{Br})_{old} - \frac{1}{y_{rk}}(z_k - c_k)$

$= 0 + \frac{c_k - z_k}{y_{rk}}$

$c_k - z_k < 0$ & $y_{rk} > 0$

$\rightarrow (z_{Br} - c_{Br})_{new} < 0$

$\therefore$ X selected

BV에 대응하는 C_B는 무조건 0으로 정리하고 시작

[Initial BFS] RHS 양수를 유지하기 위해 X_A 도입해야할수도.

① Two-Phase Method

Phase 1 : Min $1x_a$ → basis 빠져나가면 삭제.

if $x_a \neq 0$, 원문제 is infeasible

if $x_a = 0$, out of basis → 삭제

if $x_a = 0$, inside basis → pivot 한번 더 or 조심해서 phase2

Phase 2 : 밑에 테이블은 유지하되 z-row만 원문제로 바꾸기.

BV에 대응되는 C_B들은 다 0으로 pivoting 하고 시작

② Big-M Method

목적함수에 $+Ma$ (min문제) 추가.

→ initial tableau 세팅 조심. z row $C_B = 0$ 되도록.

if $x_a > 0$: infeasible   * pivoting 하다가 basis 나가면 삭제

if $x_a = 0$ (NBV) : 끝

if $x_a = 0$ (BV) : 끝

[Revised Simplex Method] • a_j, C_B 구해두기!!!

• 구성.

	Basic Inverse	RHS
z	$w = C_B B^{-1}$	$C_B b$
X_B	B^{-1}	$B^{-1}b$

* C_B 넣는거 조심. big-M 있으면 M으로 계산해야됨.

- $z_j - c_j = wa_j - c_j$ 계산해서 x_k (entering 결정)
- $y_k = B^{-1}a_k$ 계산
- RHS/y_k 해서 MRT 진행. leaving 결정.

	Basic Inverse	RHS	x_k
z	$w = C_B B^{-1}$	$C_B b$	$z_k - c_k$
X_B	B^{-1}	$B^{-1}b$	y_k

→ y_{rk} 기준으로 pivot 해서 x_k 대입해주기. 계산 조심...

Simplex Method for Bounded Var.

$z = z_0 - \sum (z_j - c_j) x_j$

$X_{BV} = B^{-1}b - y \cdot \Delta$

optimal ⟶ if not (enter X_{NBV})

	optimal	if not (enter X_{NBV})
$x_{NBV} = l$	$z_j - c_j \leq 0$	$z_j - c_j > 0$
$x_{NBV} = u$	$z_j - c_j \geq 0$	$z_j - c_j < 0$

Φ max 선택 → $x_{NBV}\uparrow$ (to u): x_{BV} $\begin{cases} y>0.\ x_{NBV}\uparrow.\ x_{BV}\downarrow : \frac{\bar{b}-l}{y} \\ y<0.\ x_{NBV}\uparrow.\ x_{BV}\uparrow : \frac{u-\bar{b}}{-y} \end{cases}$

→ $x_{NBV}\downarrow$ (to l): x_{BV} $\begin{cases} y>0.\ x_{NBV}\downarrow.\ x_{BV}\uparrow : \frac{u-\bar{b}}{-y} \\ y<0.\ x_{NBV}\downarrow.\ x_{BV}\downarrow : \frac{\bar{b}-l}{y} \end{cases}$

$\Delta = \min \left\{ \begin{array}{l} u_{NBV} - l_{NBV} \\ \text{for each } X_{BV} \text{ y에 따라 } \left[\frac{\bar{b}-l}{y} \text{ or } \frac{u-\bar{b}}{-y} \right] \end{array} \right\}$

↳ leaving X_{BV} 결정.

$z(RHS) = 숫 - (z-c)_{NBV} \cdot \Delta$

$[X^{전}_{BV}] = [B^{-1}b] - [y] \cdot \Delta = [업데이트]$ → leave to l or u : NBV

$X^{전}_{NBV} = \begin{cases} l + \Delta \\ u - \Delta \end{cases}$ → BV로 변환.

→ pivot 으로 iteration 마무리 (l, u 잘 정돈) (RHS 건들지 말기)

※ ① initial BFS 조심 ② min / max 조심 ③ Phase 1 → Phase 2 넘어갈때 RHS 세팅 조심. ④ alternative 조심

* degeneracy 있으면 무조건 cycling X. → M.R.T 에서 tie 있으면

↳ cycling 있을수도 있음.

① Lexicographic Rule for Exiting

$$I_0 = \left\{ r : \frac{\bar{b}_r}{y_{rk}} = \text{minimum}_{1 \leq i \leq m} \left\{ \frac{\bar{b}_i}{y_{ik}} : y_{ik} > 0 \right\} \right\}.$$

ton, namely $I_0 = \{r\}$, then x_{B_r} leaves the basis. C

$$I_1 = \left\{ r : \frac{y_{r1}}{y_{rk}} = \text{minimum}_{i \in I_0} \left\{ \frac{y_{i1}}{y_{ik}} \right\} \right\}.$$

n, namely, $I_1 = \{r\}$, then x_{B_r} leaves the basis. Ot

eneral, I_j is formed from I_{j-1} as follows:

$$I_j = \left\{ r : \frac{y_{rj}}{y_{rk}} = \text{minimum}_{i \in I_{j-1}} \left\{ \frac{y_{ij}}{y_{ik}} \right\} \right\}.$$

② Bland's Rule

- enter : $z-c > 0$ 인것들 중 가장 작은 인덱스
- leave : tie 중 작은 인덱스